

Chenchen Mao

RESEARCH SCIENTIST | HUMAN-CENTERED AI, EXPERIMENTATION & MEASUREMENT

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PROFILE

Ph.D. candidate in Computer Science studying how people perceive, trust, and disclose to general-purpose AI. Runs controlled and in-situ field experiments, builds the research instrumentation they require — browser extensions and experimental platforms — and models the results with structural equation modeling. Particular focus on measurement validity: establishing that an instrument measures what it claims before drawing conclusions from it.

EDUCATION

Ph.D., Computer Science — Lehigh University, Bethlehem, PA | Expected Dec 2026 | Candidacy completed

Dissertation: From Tool to Social Actor — Capability Perception, Anthropomorphism, and Differentiated Trust in General-Purpose AI

M.S., Computer Science — Syracuse University | 2021 **B.S., Computer Science & Mathematics** — University of Denver | 2018

SKILLS

Research methods: Behavioral and in-situ field experiments; survey design; structural equation modeling (lavaan); mediation and moderation; measurement validation (CFA, discriminant validity); multiplicity control (Benjamini-Hochberg); regression; ANOVA / ART ANOVA; equivalence testing (TOST); qualitative coding with intercoder reliability; Qualtrics, Prolific

Programming & systems: Python, R (lavaan), JavaScript, Node.js / Express, MongoDB, MySQL (familiarity), Git, Jupyter; Chrome extension development, HTML / CSS / DOM manipulation, experimental platform development

AI / LLM systems: LLM API integration, retrieval-grounded generation, conversational memory design, prompt design, automated text-quality evaluation (BERTScore, COMET, readability metrics)

Languages: Chinese (native), English (fluent)

RESEARCH EXPERIENCE

Graduate Research Assistant — Lehigh University, Bethlehem, PA | Jan 2023 – Present

Outer Limits: controlled content manipulation within the Reddit interface (N = 219) **Accepted at CSSSA 2026**

- Built and released an open-source Chrome extension and Express/MongoDB backend that replaces post and comment content inside the live Old Reddit interface, logs interaction events, and contains every participant action so nothing reaches the host platform.
- **Why it matters:** independent researchers previously had to choose between ecological validity (real platforms, no experimental control) and control (simulated platforms of uncertain generalizability). This removes that trade-off without requiring platform-company cooperation or privileged data access.
- Validated the instrument rather than assuming it: ART ANOVAs found no significant effects, exploratory TOSTs met a $d = \pm 0.50$ equivalence bound on every measure, and qualitative coding (seven themes, Krippendorff's $\alpha > .80$) converged — establishing that the manipulation is not detectable as an artifact.

How Far Do Capability Cues Travel? Anthropomorphism and Differentiated Trust (N = 270)

- Designed and deployed a randomized 2x2 field experiment (functional breadth x explanation) in situ on Reddit through Outer Limits, measuring six latent outcomes under Benjamini-Hochberg control.
- **Finding: capability cues do not travel.** The functional-breadth display moved perceived breadth ($\beta = .402$) and was the only randomized main effect to survive correction; an explanatory rationale had no reliable effect despite 75% recognition. Feature recognition is therefore not evidence that an interface intervention succeeded — evaluations must measure the specific judgment a feature is meant to change.
- **Finding: trust must be measured as separate dimensions.** Perceived intelligence related to ability, integrity, and benevolence, whereas the incremental association of anthropomorphism was specific to benevolence — established by Wald comparisons and replicated under WLSMV. A single global trust score conceals which judgment actually moved.

When Readability and Source Retention Diverge: An Evaluability Gap in AI Translation (N = 306)

- Ran a focal 2x2 comparison (source-text condition x output rendering) on TransLingo, a plain-text translation interface built for the study, with a stimulus audit (Δ Flesch, Jaccard, BERTScore, COMET) documenting that fidelity-oriented outputs retained more source content.
- **Finding: source visibility is not source evaluability.** Every participant saw source and output side by side, yet holistic quality ratings tracked the audited retention difference only for simple texts; for complex prose, outputs differing sharply in source overlap (Jaccard .138 vs. .452) received nearly identical ratings.

- **Why it matters for AI evaluation:** holistic human preference ratings are the signal behind RLHF and DPO. A rating that does not register fidelity loss cannot supply a penalty for it — this identifies a condition under which human feedback systematically fails to constrain a model.
- Modeled the appraisal structure with SEM in R (CFI = .940, RMSEA = .058): perceived quality → perceived intelligence → agency-oriented anthropomorphism → task-performance trust → disclosure willingness across six personal-information domains, with bootstrap indirect effects and FDR correction.

SELECTED PROJECTS

CellarEcho | *end-to-end LLM application, sole builder* [Node.js/Express · MongoDB · LLM APIs · Render]

- Designed, built, and deployed an LLM application answering inventory-specific questions, with third-party API integration, database design, authentication, and search-based grounding to reduce unsupported answers; added a memory layer that summarizes prior sessions and injects history, user facts, and retrieval signals into generation.

AI Truman | *social media simulation platform*

- Took over development from a departing developer and restructured the experimental design and data model to support feed- and post-level analysis of AI content, attention, engagement, and platform trust.

PUBLICATIONS

Peer-Reviewed and Accepted

- **Mao, C.**, Shi, H., Jia, H., Unhuryan, D., Baumer, E. P. S., & DiFranzo, D. (2026). Outer Limits: An experimental approach to controlled content manipulation within the Reddit interface. *Computational Social Science Society of the Americas (CSSSA)*. **(Accepted)** arXiv:2608.10115
- Lazard, A. J., DiFranzo, D., Nicolla, S., Aghajari, Z., Pulido, M., **Mao, C.**, Vereen, R. N., & Dasgupta, N. (2025). Sharing antivaping social media messages. *Journal of Adolescent Health*, 77(2), 253–261.
- Zhong, W., Suk, J., Pruden, M. L., Yang, Y., Zhang, Y., **Mao, C.**, & DiFranzo, D. (2026). Rethinking digital trace inference in political communication: Evidence from a simulated platform experiment. *Journal of Information Technology & Politics*. **(Accepted)**
- Nicolla, S., McKenzie, A. H., Lazard, A. J., DiFranzo, D., Aghajari, Z., **Mao, C.**, & Dasgupta, N. (2025). Young people are willing to intervene against rape myths online: A simulated social media experiment. *Journal of Adolescent Health*.

Preprints and Manuscripts

- **Mao, C.**, Shi, H., Jia, H., Wegrzyn, E., & DiFranzo, D. (2026). When readability and source retention diverge: An evaluability gap in AI translation. arXiv:2608.19083
- **Mao, C.**, et al. (2026). How far do capability cues travel? Anthropomorphism and differentiated trust in a platform-embedded AI assistant.

TEACHING EXPERIENCE

Teaching Assistant, Computer Science Department — Lehigh University | Aug 2021 – Dec 2022

- Supported software engineering and introductory programming courses: assigned student teams to graders, tracked that weekly grader–team meetings took place, and handled regrade requests.

HONORS / CERTIFICATIONS

- Named to The Distinguished Dozen, *Journal of Adolescent Health* (2025), as co-author on “Sharing antivaping social media messages.”
- WSET Level 3 Award in Wines, certified Jan 2025.